FEUP

FACULTY OF ENGINEERING - UNIVERSITY OF PORTO

Systems Engineering

Risk Management

SWAID - Plate Resonance (PR)

PLATE RESONANCE
TEAM

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1 Introduction

1.1 Document Presentation

This document constitutes the Risk Management deliverable for the Systems Engineering course, prepared in support of the mid-term review presentation. Its purpose is to detail the comprehensive risk identification, analysis, and mitigation strategies for the Plate Resonance (PR) subsystem, integrated within the global SWAID project.

1.2 Terms and Definitions

SWAID Standing Waves Acoustic Interactive Demonstrator.

PR Plate Resonance (Plate Resonance Subsystem).

CP Chladni Plate.

HI Human Interface (Human Interface Subsystem).

1.3 Changes

Version	Date	Author	Description
1.0	2026-02-27	Lucas Aparício	Template creation
2.0	2026-04-20	Lucas Aparicio	First Version
2.1	2026-04-24	Lucas Aparicio	Final Version

2 Risk Management Methodology

To ensure the successful integration and operation of the Plate Resonance (PR) subsystem, the team has adopted a structured risk management approach based on the Project Management Body of Knowledge (PMBOK) framework. This establishes a continuous, iterative lifecycle designed to identify, evaluate, treat, and monitor potential threats to the project's performance, schedule, and safety requirements.

Ground Rules and Assumptions

- External physical components provided by the CP team (structural enclosure, Chladni plate) will be delivered on schedule to allow for adequate integration testing.
- The HI team will adhere to the agreed-upon ZeroMQ JSON interface for inter-process communication and transmit valid, expected data.

Risk Categories

- **Integration:** Risks arising from physical or software interfaces between subsystems, including mismatched communication protocols, electrical boundary issues, and hardware handoffs.
- **Development:** Internal technical risks related to the PR software architecture, algorithm performance, memory management, and real-time execution deadlines.
- **Management:** Project-level risks involving schedule delays, scope creep, resource availability, and uncommunicated specification changes between teams.

Risk Management Process
The execution of this methodology follows four sequential phases:

1. **Identification:** Brainstorming technical constraints and dependencies to catalog potential sources of failure within the defined risk categories.
2. **Analysis:** Evaluating each identified risk based on Probability (P) and Impact (I) on a 1-5 scale. A Risk Score ($P \times I$) is calculated to prioritize threats.
3. **Response:** Developing specific architectural prevention and fallback strategies for Very High risks (Score $\geq$ 15).
4. **Monitoring:** Tracking the status of known risks and evaluating the effectiveness of implemented responses during weekly team meetings.

3 Risk Identification

Based on previous deliverables and the limitations by them implemented, the team conducted a comprehensive brainstorming session using the established risk categories. The resulting risks are formally cataloged below to facilitate targeted qualitative analysis.

PR-R01 Acoustic-to-Lighting Sync Jitter: Inconsistencies or latency between the audio signal generation and the corresponding lighting system triggers. This is a risk because it breaks the perceptual illusion for the user, making the installation feel disconnected and unpolished.

PR-R02 Loss of Real-Time Perception: The system response to a user's gesture takes too long to process and reflect on the plate. It destroys the interactive nature of the project; if users do not feel they are directly controlling the system, the primary goal fails.

PR-R03 CP & HI Development Delay: The Chladni Plate or Human Interaction teams falling behind their scheduled milestones. This presents a major risk as it creates a bottleneck, preventing our team from performing necessary cross system integration and testing before the final deadline.

PR-R04 Resonance Dictionary Drift: The mapped Chladni resonance frequencies shift over time due to environmental factors, e.g. the plate's physical properties. This presents a risk because the software will output "correct" frequencies that no longer create the intended geometric patterns in the physical world.

PR-R05 Sound Card channel control limitation (De-risked): The inability to independently control phase, amplitude, and frequency across different soundcard output channels. Originally a critical risk that would prevent complex node steering on the plate, this has been completely resolved via custom ALSA drivers and verified on an oscilloscope.

PR-R06 OS Scheduling Jitter: The standard Linux operating system preempting our real time audio thread to execute background tasks. This is a risk because it causes buffer underruns and missing the strict 20ms constraint, leading to audio dropouts and analog signal distortion.

PR-R07 Informal Interface Specifications: Operating without a formal interface control document with the CP team. This presents a risk of mismatched assumptions regarding electrical boundaries, which could lead to unusable signals or hardware damage upon integration.

PR-R08 Uncommunicated Specification Changes: Other teams altering their hardware or data requirements without immediately notifying our team. Last minute software rewrites, highly increasing the likelihood of introducing bugs right before the deadline.

PR-R09 Blocked Integration Testing: Unavailability of the physical amplifier and plate setup from the CP team when needed. This is a risk because it leaves our team unable to validate the analog signals in a real-world physical environment.

PR-R10 Long-Term Synchronization Drift: Minute clock inaccuracies causing the generated analog channels to slowly fall out of phase with one another. This risks degrading the stability and clarity of the Chladni patterns over hours of continuous operation, requiring frequent system resets.

PR-R11 Scope Creep: Attempting to add unplanned, non-essential features (complex UI elements or extra modulations, etc.). This presents a risk as it consumes valuable development time and introduces unnecessary instability into the core real time processing loop.

PR-R12 Insufficient Transducer Force: The 25W transducers lacking the mechanical power to properly displace sand on the stainless steel plate. Without enough force, the high-frequency nodes will not be visible, failing the project's primary visual objective.

PR-R13 Simulation Divergence (Edge Dampening): Real world physical constraints, such as the perimeter rim dampening the plate's vibrations, causing the plate to behave differently than what was predict. This risks rendering our calculated frequencies useless, as they won't form stable patterns on the actual physical hardware.

PR-R14 System Crash on Rapid HI Input: The ALSA generation thread failing under a flood of continuous, rapid data from user gestures. A risk because it requires a full system reboot, disrupting the user experience and violating the requirement for continuous operation.

PR-R15 Validation Failure (Sand Settling Time): The physical sand taking more than the 60-second metric window to settle into a new pattern after a frequency change. First time users will fail to successfully trigger and visually comprehend a pattern change in the designated time.

PR-R16 Incorrect HI Data Transmission: The HI gesture software transmitting out-of-bounds numerical values or undetermined symbols to the audio engine. Processing this unfiltered data risks crashing the real-time ALSA thread or outputting uncontrolled analog signals that permanently damage the physical amplifiers.

PR-R17 Critical Component Disconnection/Failure: The sudden failure or physical disconnection of a critical hardware component during an active exhibition. This presents a severe risk because a single hardware fault breaks the complete interactive loop; without a robust degradation protocol, the entire installation would appear dead and unengaging to the public.

4 Qualitative Risk Analysis and Impact Matrix

The identified risks were evaluated on a 1-5 scale for both Probability (P) and Impact (I). The resulting Risk Score ($P \times I$) dictates the priority level of each threat. Risks achieving a score of 15 or higher are classified as High priority and require mandatory, immediate mitigation planning.

5 Risk Response and Mitigation Strategies

In accordance with our methodology, risks identified as **Very High** or **Extreme** require dedicated, formalized action plans. The following strategies outline how our team will prevent these risks from occurring, alongside fallback mitigations if the preventive measures fail.

Table 1: Qualitative Risk Analysis and Impact Matrix

ID	Risk Description	Category	P	I	Score	Level
PR-R01	Acoustic-to-Lighting Sync Jitter	Integration	3	5	**15**	Very High
PR-R02	Loss of Real-Time Perception	Integration	3	5	**15**	Very High
PR-R03	CP & HI Development Delay	Management	2	5	**10**	High
PR-R04	Resonance Dictionary Drift	Development	4	2	**8**	Medium
PR-R05	Sound Card channel control limitation	Development	0	5	**0**	Derisked
PR-R06	OS Scheduling Jitter (20ms Deadline)	Development	3	4	**12**	High
PR-R07	Informal Interface Specifications	Management	3	4	**12**	High
PR-R08	Uncommunicated Specification Changes	Management	2	4	**8**	Medium
PR-R09	Blocked Integration Testing	Management	3	4	**12**	High
PR-R10	Long-Term Synchronization Drift	Development	2	3	**6**	Medium
PR-R11	Scope Creep	Management	2	2	**4**	Low
PR-R12	Insufficient Transducer Force	Hardware	4	4	**16**	Very High
PR-R13	Simulation Divergence (Edge Dampening)	Hardware	3	4	**12**	High
PR-R14	System Crash on Rapid HI Input	Integration	2	5	**10**	High
PR-R15	Validation Failure (Sand Settling Time)	Integration	3	4	**12**	High
PR-R16	Incorrect HI Data Transmission	Integration	3	5	**15**	Very High
PR-R17	Critical Component Disconnection/Failure	Hardware	3	5	**15**	Very High

5.1 Mitigation Plans for Very High Risks

PR-R01 Acoustic-to-Lighting Sync Jitter
Prevention Plan: The system architecture will utilize a sequenced callback mechanism. Because the mechanical formation of the acoustic pattern on the plate is significantly slower than digital execution, the audio thread will initiate the frequency change first. Only after accommodating the mechanical transition period will the system send a "Pattern Ready" callback to the LED controller. This ensures the instantaneous lighting transition strictly aligns with the moment the physical pattern is established.
Fallback Protocol: If the dynamic callback mechanism proves unreliable or fails to synchronize accurately, the team will revert to a static, empirically derived timing model. This will require dedicated lab sessions to physically measure and record the exact mechanical formation time for every individual Chladni pattern in the dictionary. These custom, pattern specific temporal offsets will then be hardcoded into the LED transition logic, ensuring visual synchronization through fixed delays rather than relying on inter-thread communication.

PR-R02 Loss of Real-Time Perception
Prevention Plan: The software architecture will enforce strict multithreading. The audio generation loop (which must meet the 20ms deadline) will run on an isolated, high-priority CPU core, while the parsing of Human Interaction (HI) gesture data will run on a separate thread. This ensures that heavy gesture math never bottlenecks the immediate acoustic response.
Fallback Protocol: If the physics calculation for a complex new Chladni pattern drops frames, the system will trigger a transitional sound effect coupled with a synchronized LED visual cue immediately upon gesture detection. This provides the user with instant audiovisual feedback while the heavier geometric calculations catch up in the background.

PR-R12 Insufficient Transducer Force
Prevention Plan: Mathematical simulations cannot guarantee real-world physical displacement. Therefore, the team will conduct an early-stage mechanical validation test utilizing the

exact 300x200x1mm stainless steel plate and the 25W transducers prior to final system assembly.
Fallback Protocol: If the 25W mechanical force proves insufficient to generate visible nodes, particularly for high-frequency patterns, the team will execute one of two contingencies: (1) utilize emergency budget reserves to procure higher wattage (e.g., 50W) transducers, or (2) restrict the software dictionary to lower-order resonant modes. While these lower-frequency modes exhibit reduced spatial complexity (fewer antinodes), they reliably demonstrate the physical phenomena while requiring less mechanical drive.

PR-R16 Incorrect HI Data Transmission
Prevention Plan: Implement a strict data sanitization and validation layer at the input parser. Before HI data reaches the core audio engine, it must be validated against a predefined safe operational range. If numerical values are transmitted that exceed the maximum or fall below the minimum safe thresholds, the sanitization layer will automatically pull or push them to the nearest safe boundary . As for if any data containing corrupted or undetermined symbols, or an unexpected size or type, it will be dropped.
Fallback Protocol: If the parser drops data continuously for more than 3 seconds due to corruption or invalidity, the system will execute two simultaneous actions: (1) it will send a critical callback alert to the operator console identifying a data stream fault, and (2) the audio engine will enter a "Safe Sequence State," generating a predefined, visually stable carrossel of Chladni patterns. The system will remain in this protective loop until a mathematically valid data packet is successfully received, at which point normal interactive operation will seamlessly resume.

PR-R17 Critical Component Disconnection/Failure
Prevention Plan: While the physical mounting and thermal management of the hardware fall under the CP team's responsibility, the joint system prevention strategy mandates the use of secure, locking connectors and strain relief mechanisms for all cabling. Furthermore, the CP team must ensure components are operated strictly within their thermal limits to prevent spontaneous burnout.
Fallback Protocol: If a subsystem completely fails, the PR software will send a diagnostic alert to the operator console specifying exactly which component was lost. Simultaneously, the system will trigger a "Standalone Exhibition Mode," dynamically adjusting its behavior based on the surviving hardware:

- **HI Camera Fails (Plate + LEDs survive):** Interactive input is lost. The PR engine will automatically loop a pre-programmed sequence of visually striking resonant frequencies, allowing the plate and LEDs to animate autonomously as a kinetic sculpture.
- **Transducers Fail (Camera + LEDs survive):** Acoustic output is lost. The PR system will bypass the audio generation loop and map the HI gesture data directly to the LED controller, showing the designated patterns but only on the LED for the time being (visual only interaction).
- **LEDs Fail (Camera + Plate survive):** Lighting is lost. The PR system will continue normal operation, translating user gestures into audio frequencies to drive the sand patterns on the plate (acoustic only interaction).
- **Dual Failure (Only one output survives):** If the camera and one output fail simultaneously, the single remaining operational output (either the plate or the LEDs) will default to a safe, pre-programmed standalone loop to maintain a baseline level of public display while the operator intervenes.

6 Risk Monitoring and Control

To ensure mitigation strategies remain effective, the Risk Register (full impact matrix) is maintained as a living document. During the weekly sessions, the team will formally evaluate whether existing risks have escalated in probability/impact, verify the implementation status of ”Very High” prevention strategies, and catalog any newly discovered integration threats.

7 Independent Subsystem Testing

To mitigate cross-team dependencies, the PR subsystem is fully testable in isolation. For inputs, automated scripts bypass the HI camera by injecting simulated ZeroMQ JSON payloads, allowing stress testing of the parser, rate limits, and the 20ms audio thread. These scripts inject malformed JSON to validate data dropping protocols, while execution time is strictly profiled. For outputs, ALSA soundcard signals are routed directly to an oscilloscope, bypassing the CP hardware. Simultaneously, the physical LED controller remains connected to verify event driven lighting callbacks, guaranteeing the subsystem is mathematically, electrically, and visually validated prior to final integration.

8 Conclusion

The Plate Resonance (PR) subsystem involves complex intersections of real-time software, physical acoustics, and cross-team dependencies. By proactively adopting this PMBOK-aligned Risk Management framework, e have identified and quantified 17 potential threats to the project’s success. Critically, we have established robust, architectural-level prevention and fallback protocols for the highest-priority risks, ensuring that our strict 20ms execution deadlines and physical node-generation goals remain protected. Moving forward, this risk register will act as a living document. Through continuous weekly monitoring and strict adherence to our mitigation strategies, the team is fully equipped to navigate technical uncertainties and deliver a stable, highly responsive acoustic subsystem for the SWAID project.